Tegkirat Kaur

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Patiala, Punjab, India

Education

Thapar Institute of Engineering and Technology - BTech. CSE - CGPA 8.82/10 **2025 – Present**
Patiala, Punjab, India

Delhi Public School - Non Medical - 86.4% **2023 – 2025**
Patiala, Punjab, India

AI Engineering & Technical Experience

Brepline | ***Software Engineer*** **2026 – Present**

- **Developing a 7.9B+ parameter large multimodal model (LMM) for AI-driven 3D modeling, generating parametric CAD models in STEP/B-Rep formats from natural-language prompts and sketches.**
- **Integrating AI with SolidWorks through custom plugins for AI-assisted CAD design, parameter editing, and automated engineering workflows.**
- **brepline.vercel.app**

PREG AI | ***SIH2026 Project — HARDWARE*** **2026**

- **PregAI is a low-cost, non-invasive EHG-based maternal monitoring system for AI-assisted preterm-birth risk assessment. It captures uterine electrical activity through abdominal electrodes, amplifies and digitizes the signal using AD620 and ADS1115, and processes it through ESP32-S3. EHG features and signal-quality metrics are transmitted via Wi-Fi to Firebase and visualized on a web dashboard. Machine-learning models are being developed using public EHG datasets to identify patterns associated with term and preterm pregnancies.**

Sitee | ***Software Engineer*** **2025**

- **Designed and developed an AI-powered application leveraging a custom 1.2B+ parameter LLM trained on IITD Cluster and IUCAA Pegasus Cluster to autonomously generate full-stack websites, streamlining developer workflows with a brutalist-inspired UI.**
- **Integrated vector databases (FAISS/Pinecone) for efficient embedding storage and retrieval, ensuring scalable AI workflows and rapid prompt execution.**
- **www.sitee.in**

Research Projects

1. **SunFLAAR: an open-source Python package for comprehensive solar flare analysis.** **(2026)**
   - **SunFLAAR is an open-source Python package for comprehensive solar flare analysis, integrating deep learning and statistical methods. It provides automated tools for multivariate time-series forecasting, active region classification, and imbalanced data metrics to aid space weather research.**
   - ***Authors: Tegkirat Kaur,* & *Jashanpreet Singh Dingra***
     https://sunflaar.vercel.app

Achievements

Hackathons

- **Smart India Hackathon** - Internal Round Clear **(2026)**
- **Niohack by NioGraph Inc. USA** - Runner Up **(2026)**
- **AlgOlympia, Chandigarh University** - Winner **(2026)**

Technical Skills

Programming Languages: **Python, C++, C, FORTAN, MERN, Bash**

Database Systems: **PostgreSQL, MySQL, MongoDB, Firebase SDK**

Cloud: **AWS, GCP, Git, Linux**

Tech Stack: **FastAPI, LangChain, LangGraph, REST API, Weights & Biases, MLflow**

Libraries / Frameworks: **PyTorch, TensorFlow, JAX, Scikit-learn, HuggingFace, OpenCV, PIL**

Machine Learning: **Transformers, GNNs, State Space Models (Mamba), RAG, LLM Fine-tuning (SFT, LoRA, QLoRA, PEFT), Agentic AI**

Visualization: **Matplotlib, Seaborn, Plotly, Power BI, Tableau**

Core CS Fundamentals: **Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Operating System**